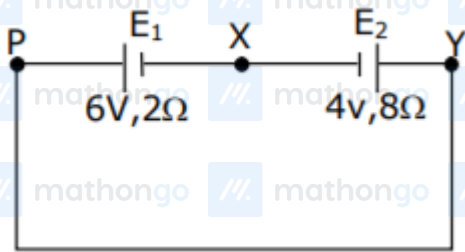


Questions with Answer Keys

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Q1: 24 Feb (Shift 1) - Single Correct

A cell E_1 of emf 6 V and internal resistance 2Ω is connected with another cell E_2 of emf 4 V and internal resistance 8Ω (as shown in the figure).



The potential difference across points X and Y is -

- (1) 3.6 V
- (2) 10.0 V
- (3) 5.6 V
- (4) 2.0 V

Q2: 24 Feb (Shift 1) - Single Correct

A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$ where $\alpha_0 = 20 \text{ A/s}$ and $\beta = 8 \text{ As}^{-2}$. Find the charge crossed through a section of the wire in 15 s.

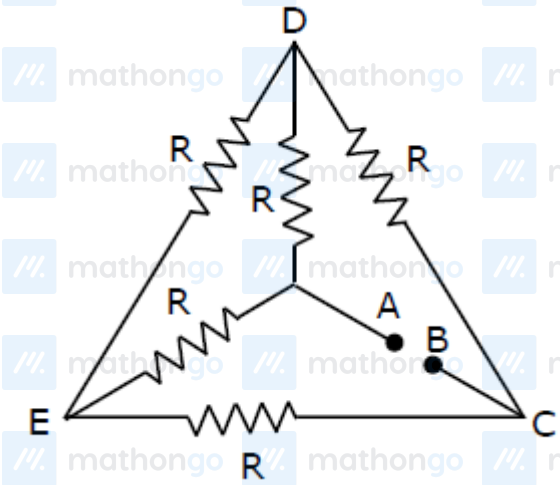
- (1) 2100C
- (2) 260C
- (3) 2250C
- (4) 11250C

Q3: 24 Feb (Shift 2) - Numerical

A cylindrical wire of radius 0.5 mm and conductivity $5 \times 10^7 \text{ S/m}$ is subjected to an electric field of 10mV/m. The expected value of current in the wire will be $x^3 \pi \text{ mA}$. The value of x is

Q4: 26 Feb (Shift 1) - Single Correct

Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is :



(1) $\frac{3R}{2}$

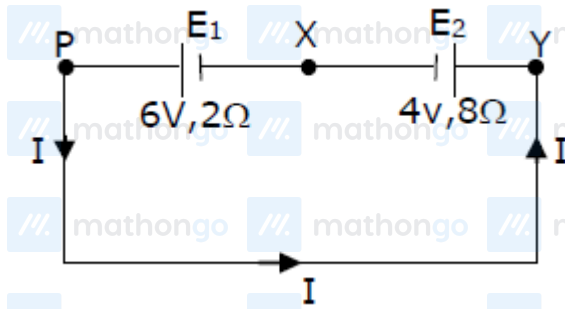
(2) $\frac{R}{2}$

(3) R

(4) $2R$

Answer Key**Q1 (3)****Q2 (4)****Q3 (5)****Q4 (3)**

Q1 (3)



$$\text{emf of } E_1 = 6\text{V}$$

$$r_1 = 2\Omega$$

$$\text{emf of } E_2 = 4\text{V}$$

$$r_2 = 8\Omega$$

$$|v_x - v_y| = \text{potential difference across points } x \text{ and } y$$

$$E_{\text{eff}} = 6 - 4 = 2\text{ V}$$

$$R_{\text{eq}} = 2 + 8 = 10\Omega$$

So, current in the circuit will be

$$\Rightarrow I = \frac{E_{\text{eff}}}{R_{\text{eq}}} \Rightarrow I = \frac{2}{10} = 0.2\text{ A}$$

Now, potential difference across points X and Y is $|v_x - v_y| = E + iR$

$$\Rightarrow |v_x - v_y| = 4 + 0.2 \times 8 = 5.6\text{ V}$$

$$\Rightarrow |v_x - v_y| = 5.6\text{V}$$

Q2 (4)

$$\text{given : } i = \alpha_0 t + \beta t^2$$

$$\alpha = 20\text{ A/s and } \beta = 8\text{ As}^{-2}$$

$$t = 15\text{sec}$$

$$\text{we know that, } i = \frac{dq}{dt} \Rightarrow \int_0^t i dt = \int_0^Q dq$$

$$\Rightarrow \int_0^{15} (\alpha_0 t + \beta t^2) dt = \int_0^Q dq$$

$$\Rightarrow Q = \left[\frac{\alpha_0 t^2}{2} + \frac{\beta t^3}{3} \right]_0^{15}$$

Hints and Solutions

MathonGo

$$\Rightarrow Q = \frac{20 \times 15 \times 15}{2} + \frac{8 \times 15 \times 15 \times 15}{3} - 0$$

$$\Rightarrow Q = 11250c$$

Q3 (5)

We know that $J = \sigma E$

$$\Rightarrow J = 5 \times 10^7 \times 10 \times 10^{-3}$$

$$\Rightarrow J = 50 \times 10^4 \text{ A/m}^2$$

Current flowing $I = J \times \pi R^2$

$$I = 50 \times 10^4 \times \pi (0.5 \times 10^{-3})^2$$

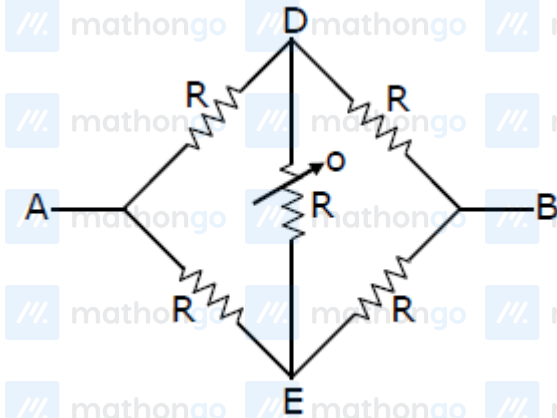
$$I = 5 \times 10^4 \times \pi \times 0.25 \times 10^{-6}$$

$$I = 125 \times 10^{-3} \pi$$

$$X = 5$$

Q4 (3)

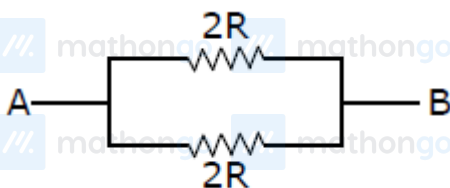
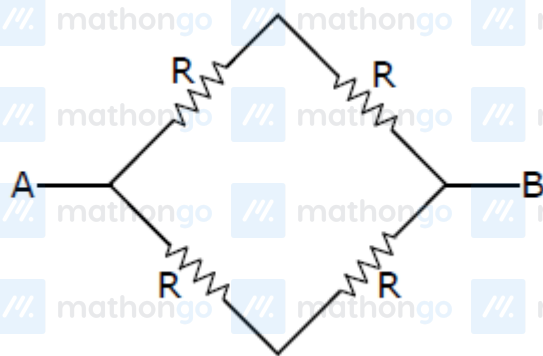
It is balanced wheat stone bridge



So, we know that

$$R_1 R_4 = R_2 R_3$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$



$$\begin{aligned} R_{eq} &= \frac{2R \times 2R}{2R + 2R} \\ &= \frac{4R^2}{4R} \Rightarrow R \end{aligned}$$